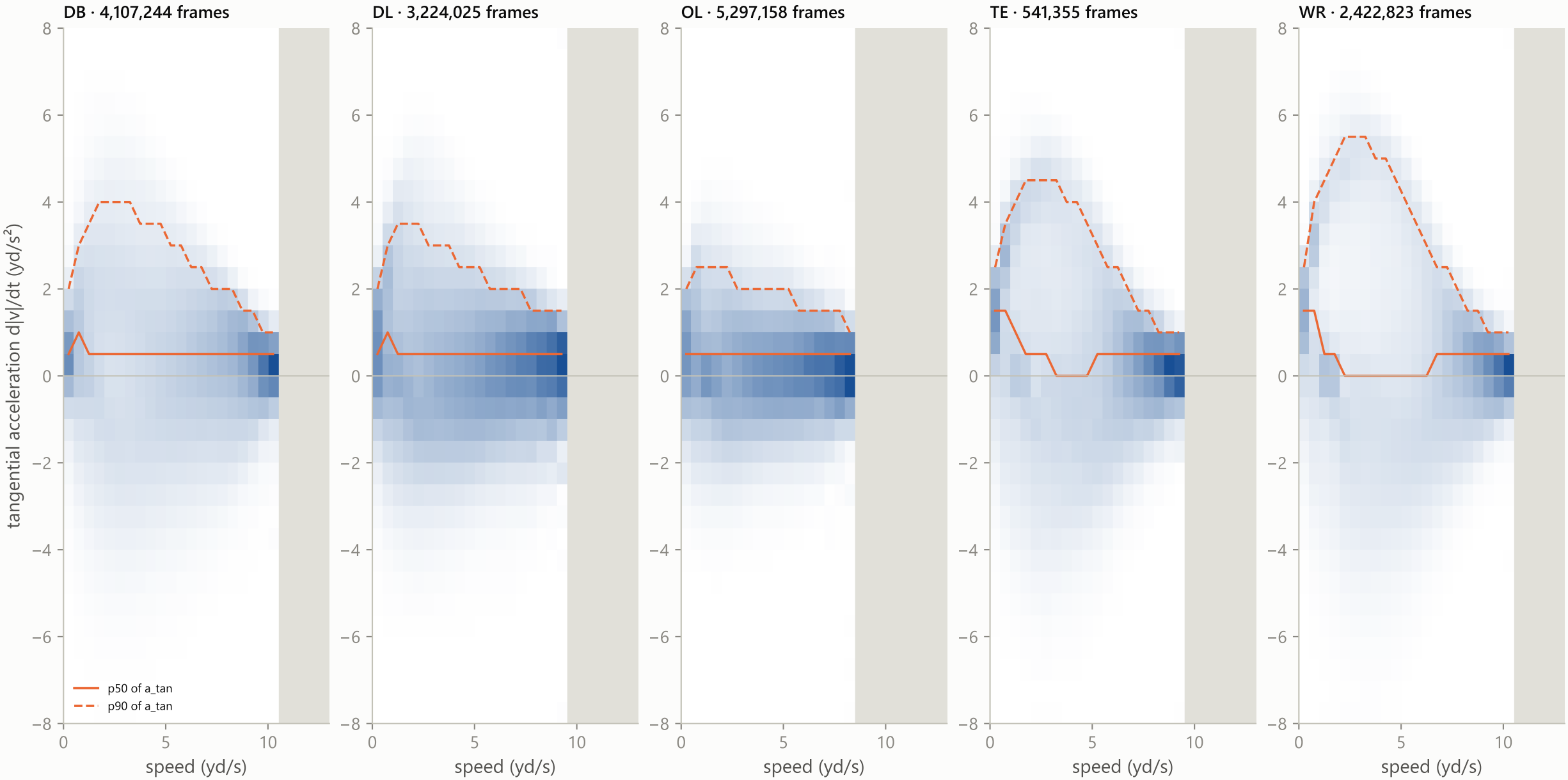
Speed–acceleration phase space by position group



Unit: post-snap frame of an in-scope athlete (a frame is NOT an independent observation; the count is a density, not a sample size). Colour: share of the frames of a 0.5 yd/s speed bin that fall in each 0.5 yd/s² acceleration bin (bins with < 200 frames are grey; accelerations beyond ± 8 yd/s² are in the end bins). Lines: median and 90th percentile of the acceleration at that speed (upper bin edge). This is the OBSERVED movement distribution of a position in its role, not a physical capability limit.